A11 Box – Wire List · United Control Box, wiring rev 0.7 · 2026-10-29 · 104 wires, 95.2 m to cut + 4 cables · build-guide order

Cut = routed + 10 % + 15 cm, rounded up to 5 cm (door wires incl. 0.30 m hinge loop). **Band**: all 1 mm² wire is yellow; its function colour goes on as a band next to the number sleeve at both ends (black L / DC power, blue N, red 230 V control, dk blue +24 V / signals, bl/wh 0 V). **Ends** from / to: fer = ferrule, fer·own = own point on the distribution block, 2×+N = twin ferrule with wire N, fork(+N) = insulated fork lug (2 under one screw with N), ring = ring lug, solder = GX16 cup, crimp = HDC-HA contact, core = supply cable core.

Stock (cut / have, m): 1 mm² yellow 69.20 / 90 OK · **2.5 mm² G/Y 7.95 / 4 SHORT 3.95** · 2.5 mm² black 2.50 / 4 OK · 2.5 mm² blue 2.80 / 4 OK · 2.5 mm² brown 2.10 / 4 OK · 4 mm² G/Y 4.10 / 5 OK · 4 mm² black 3.90 / 5 OK · 4 mm² blue 1.25 / 5 OK · 6 mm² braided strap 1.40 (to buy) · **Ferrules:** 1 mm² ×77, 2.5 mm² ×10, 4 mm² ×17, twin 2 × 1 mm² ×19, twin 2 × 4 mm² ×1

⚠ **Before any wiring:** supply cord unplugged, QF0–QF4 OFF; if the VFD (spindle drive) has been powered, wait until its display and charge lamp are out. Number sleeve and colour band on **both** ends before fitting; ferrule every stranded end; max. 2 conductors per terminal (R1); tug-test each screw.

	No.	From	To	Wire mm²	Band	Cut	Ends (from / to)	Function · ⚠ check before / while wiring
1 · Protective earth – 12 wires, 13.45 m								
⚠ **CHECK FIRST:** Scrape the paint under the body and plate studs, serrated washer under the lug. PE is G/Y only, never plain yellow. After the group: every PE point beeps to the earth bar.								
☐	**138**	Earth bar 1	Earth bar 2	4 G/Y		**0.60**	ring M6 / ring M6	link (both bars sit on insulators)
☐	**139**	Enclosure body (earth stud)	Earth bar 1	4 G/Y		**0.80**	ring M6 / ring M6	NEW: the body had no PE conductor ⚠ **bare metal under the lug**
☐	**140**	Mains (gland) PE	Earth bar 1	4 G/Y		**0.65**	core / ring M6	⚠ **connect last, after § 7.1 (dead tests)**
☐	**141**	Backplate	Earth bar 1	4 G/Y		**0.45**	ring M6 / ring M6	⚠ **bare metal under the lug**
☐	**142**	Door strap	Earth bar 1	6 braided strap		**1.40**	ring M6 / ring M6	door loop
☐	**143**	EMI filter GND	Earth bar 1	4 G/Y		**1.60**	fer / ring M6	
☐	**144**	VFD PEc	Earth bar 1	2.5 G/Y		**1.15**	fer / ring M6	
☐	**145**	PSU 48V PE	Earth bar 2	2.5 G/Y		**1.30**	fork / ring M6	
☐	**146**	PSU 24V PE	Earth bar 2	2.5 G/Y		**1.15**	fork / ring M6	
☐	**147**	DIN-rail earth stud	Earth bar 2	2.5 G/Y		**1.65**	ring M6 / ring M6	
☐	**148**	PSU 24V -V2	Earth bar 2	2.5 G/Y		**1.20**	fork+169 / ring M6	PELV bond: the only 0 V to PE bond ⚠ **must beep 0 V ↔ PE in § 7.1 (dead tests)**
☐	**149**	Schuko PE	Earth bar 2	2.5 G/Y		**1.50**	fer / ring M6	NEW: missing in rev 0.5 (pump was unearthed)
2 · 230 V power and VFD – 24 wires, 19.35 m								
⚠ **CHECK FIRST:** Supply cord unplugged. L = black, N = blue; read each part's label before fitting: Z1 (EMI filter) LINE in / LOAD out, VFD (spindle drive) L / N in and U / V / W out. KM1 (contactor) switches L only. Tighten and tug-test every 230 V screw.								
☐	**101**	Mains (gland) L	Main CB 1	2.5 black		**1.50**	core / fer	Mains L ⚠ **connect the supply cable last, after the § 7.1 (dead tests) dead tests**
☐	**102**	Mains (gland) N	Main CB 3	2.5 blue		**1.55**	core / fer	Mains N ⚠ **connect last, after § 7.1 (dead tests)**
☐	**103**	Main CB 2	EMI filter LIN	4 black		**0.30**	fer / fer	⚠ **Z1 (EMI filter) LINE side (from QF0 (main breaker)); do not swap LINE / LOAD**
☐	**104**	Main CB 4	EMI filter NIN	4 blue		**0.35**	fer / fer	⚠ **Z1 (EMI filter) LINE side; do not swap LINE / LOAD**
☐	**105**	EMI filter LOUT	Contactor 1L1	4 black		**0.45**	2×+111 / fer	Unswitched L
☐	**106**	EMI filter NOUT	N busbar XN	4 blue		**0.90**	fer / fer·own	N in
☐	**109**	Contactor 2T1	L busbar XL	4 black		**0.75**	fer / fer·own	Switched L in
☐	**111**	EMI filter LOUT	CB-24V 1	4 black		**0.55**	2×+105 / fer	24 V PSU L (unswitched)
☐	**112**	CB-24V 2	PSU 24V L	1 yellow	**black**	**1.00**	2×+130 / fork	
☐	**113**	N busbar XN	PSU 24V N	1 yellow	**blue**	**0.85**	fer·own / fork	
☐	**114**	L busbar XL	CB-VFD 1	4 black		**0.60**	fer·own / fer	Switched L
☐	**115**	CB-VFD 2	VFD L	2.5 black		**1.00**	fer / fer	⚠ **VFD (spindle drive) input L, never U / V / W**
☐	**116**	N busbar XN	VFD N	2.5 blue		**1.25**	fer·own / fer	⚠ **VFD (spindle drive) input N, never U / V / W**
☐	**117**	L busbar XL	CB-48V 1	4 black		**0.60**	fer·own / fer	Switched L
☐	**118**	CB-48V 2	PSU 48V L	1 yellow	**black**	**1.15**	fer / fork	
☐	**119**	N busbar XN	PSU 48V N	1 yellow	**blue**	**0.95**	fer·own / fork	
☐	**120**	L busbar XL	CB-PUMP 1	4 black		**0.65**	fer·own / fer	Switched L
☐	**121**	CB-PUMP 2	Schuko L	1 yellow	**black**	**0.40**	fer / fer	Pump
☐	**122**	N busbar XN	Schuko N	1 yellow	**blue**	**0.45**	fer·own / fer	Pump
☐	**128**	VFD P+	Braking resistor T1	1 yellow	**black**	**1.25**	fer / ring	⚠ **keep ≥ 50 mm from the tube (very hot)**
☐	**129**	VFD PB	Braking resistor T2	1 yellow	**black**	**1.00**	fer / ring	⚠ **keep ≥ 50 mm from the tube (very hot)**
☐	**130**	CB-24V 2	K1 41 (NEMA 12)	1 yellow	**red**	**0.85**	2×+112 / fer	Coil supply L (protected by QF4 C6)
☐	**131**	K1 44 (NEMA 8)	Contactor A2	1 yellow	**red**	**0.35**	fer / fer	Switched coil L ⚠ **leave the KM1 (contactor) A2 end off until § 7.2 (power-up)**
☐	**132**	N busbar XN	Contactor A1	1 yellow	**blue**	**0.65**	fer·own / fer	Coil N
3 · 48 V drive supply – 6 wires, 6.10 m								
⚠ **CHECK FIRST:** PSU input selector (if fitted) on 230 V. Fit the drive ends only after § 7.2 (power-up) reads G1 (48 V PSU) at 46–50 V DC. R3 open on 150 / 152 / 154 (1 mm² on the 15 A PSU): fuse each drive or use 1.5–2.5 mm² first.								
☐	**150**	PSU 48V +V1	DMA860H-X AC1	1 yellow	**black**	**1.00**	fork / fer	48 V ⚠ **drive end off until G1 (48 V PSU) reads 46–50 V; R3: fuse or ≥ 1.5 mm²**
☐	**151**	PSU 48V -V1	DMA860H-X AC2	1 yellow	**black**	**0.95**	fork / fer	⚠ **drive end off until § 7.2 (power-up)**
☐	**152**	PSU 48V +V2	DMA860H-Y AC1	1 yellow	**black**	**1.10**	fork / fer	⚠ **drive end off until G1 (48 V PSU) reads 46–50 V; R3: fuse or ≥ 1.5 mm²**
☐	**153**	PSU 48V -V2	DMA860H-Y AC2	1 yellow	**black**	**1.05**	fork / fer	⚠ **drive end off until § 7.2 (power-up)**
☐	**154**	PSU 48V +V3	DMA860H-Z AC1	1 yellow	**black**	**1.00**	fork / fer	⚠ **drive end off until G1 (48 V PSU) reads 46–50 V; R3: fuse or ≥ 1.5 mm²**
☐	**155**	PSU 48V -V3	DMA860H-Z AC2	1 yellow	**black**	**1.00**	fork / fer	⚠ **drive end off until § 7.2 (power-up)**
4 · 24 V control – 32 wires, 28.45 m								
⚠ **CHECK FIRST:** Dark blue band = +24 V, blue/white = 0 V: check the polarity at every PSU and A11 (controller) terminal. Door wires go through the spiral-wrapped hinge loop. Leave 131 (KM1 A2), 166 and 168 (A11) off until § 7.2 (power-up). Sensors (NPN NO): brown +24 V, blue 0 V; slide the GX16 (round connector) shell and heat-shrink on before soldering.								
☐	**162**	PSU 24V +V1	F1 in	2.5 brown	**dk blue**	**1.10**	fork / fer	Control fuse (short, 2.5 mm²: unfused PSU side)
☐	**164**	PSU 24V +V2	F2 in	2.5 brown	**dk blue**	**1.00**	fork / fer	Signal fuse (short, 2.5 mm²: unfused PSU side)
☐	**166**	F2 out	A11 DC24V	1 yellow	**dk blue**	**1.40**	2×+168 / fer	⚠ **A11 (controller) end off until F2 (signal fuse) out reads 23–25 V DC (§ 7.2)**
☐	**167**	PSU 24V -V1	A11 DCGND	1 yellow	**bl/wh**	**0.95**	fork+207 / 2×+171	
☐	**168**	F2 out	A11 24V	1 yellow	**dk blue**	**1.40**	2×+166 / 2×+340	Sensor/input supply ⚠ **A11 (controller) end off until F2 (signal fuse) out reads 23–25 V DC**
☐	**169**	PSU 24V -V2	A11 GND	1 yellow	**bl/wh**	**1.00**	fork+148 / 2×+337	
☐	**170**	Contactor 43	Fan +	1 yellow	**dk blue**	**1.35**	2×+220 / fer	NEW: fan behind F1 (was unfused on TB24+)
☐	**171**	A11 DCGND	Fan -	1 yellow	**bl/wh**	**0.85**	2×+167 / fer	
☐	**201**	F1 out	E-stop 1	1 yellow	**dk blue**	**1.90**	2×+220 / 2×+223	Chain; door loop
☐	**202**	E-stop 2	OFF 1	1 yellow	**dk blue**	**0.35**	fer / fer	Chain; door loop

		No.	From	To	Wire mm²	Band	Cut	Ends (from / to)	Function · ⚠ check before / while wiring
66	☐	**203**	OFF 2	ON 1	1 yellow	**dk blue**	**0.25**	2×+205 / fer	Chain; door loop
67	☐	**204**	ON 2	K1 A1 (NEMA 13)	1 yellow	**dk blue**	**1.70**	fer / 2×+206	Start; door loop
68	☐	**205**	OFF 2	K1 11 (NEMA 9)	1 yellow	**dk blue**	**1.85**	2×+203 / fer	Latch; door loop
69	☐	**206**	K1 14 (NEMA 5)	K1 A1 (NEMA 13)	1 yellow	**dk blue**	**0.80**	fer / 2×+204	Latch
70	☐	**207**	PSU 24V -V1	K1 A2 (NEMA 14)	1 yellow	**bl/wh**	**0.95**	fork+167 / fer	Coil return
71	☐	**220**	F1 out	Contactor 43	1 yellow	**dk blue**	**1.05**	2×+201 / 2×+170	Yellow lamp supply
72	☐	**221**	Contactor 44	Yellow lamp +	1 yellow	**dk blue**	**1.70**	fer / fer	door loop
73	☐	**222**	K1 A2 (2nd screw) (NEMA 14)	Yellow lamp -	1 yellow	**bl/wh**	**1.75**	fer / fer	0 V to the door; door loop
74	☐	**223**	E-stop 1	Green lamp +	1 yellow	**dk blue**	**0.50**	2×+201 / 2×+225	+24 V CONTROL (door loop); door loop
75	☐	**225**	Green lamp +	Red lamp +	1 yellow	**dk blue**	**0.30**	2×+223 / fer	door loop
76	☐	**316**	A11 OGND	DMA860H-X ALM-	1 yellow	**bl/wh**	**0.75**	2×+326 / 2×+318	
77	☐	**318**	DMA860H-X ALM-	DMA860H-Y ALM-	1 yellow	**bl/wh**	**0.55**	2×+316 / 2×+320	
78	☐	**320**	DMA860H-Y ALM-	DMA860H-Z ALM-	1 yellow	**bl/wh**	**0.50**	2×+318 / fer	
79	☐	**326**	A11 OGND	VFD COM	1 yellow	**bl/wh**	**0.80**	2×+316 / fer	Output common ⚠ **§ 7.2 (power-up): VFD (spindle drive) COM ↔ G2 (24 V PSU) –V < 1 V DC**
80	☐	**337**	A11 GND	X-LIM 3	1 yellow	**bl/wh**	**0.75**	2×+169 / solder	Sensor 0 V (BU)
81	☐	**338**	X-LIM 3	Y-LIM 3	1 yellow	**bl/wh**	**0.40**	solder / solder	Sensor 0 V (loop)
82	☐	**339**	Y-LIM 3	Z-LIM 3	1 yellow	**bl/wh**	**0.40**	solder / solder	Sensor 0 V (loop)
83	☐	**340**	A11 24V	X-LIM 4	1 yellow	**dk blue**	**0.75**	2×+168 / solder	Sensor +24 V (BN)
84	☐	**341**	X-LIM 4	Y-LIM 4	1 yellow	**dk blue**	**0.40**	solder / solder	Sensor +24 V (loop)
85	☐	**342**	Y-LIM 4	Z-LIM 4	1 yellow	**dk blue**	**0.40**	solder / solder	Sensor +24 V (loop)
86	☐	**346**	Z-LIM 3	Tool setter 2	1 yellow	**bl/wh**	**0.40**	solder / solder	0 V (loop)
87	☐	**348**	Tool setter 2	Tool setter 4	1 yellow	**bl/wh**	**0.20**	solder / solder	0 V to the over-travel switch (jumper in the connector)
88	**5 · Signals – 30 wires, 27.85 m**								
89	⚠ **CHECK FIRST:** 5 V / PULSE / DIR are 5 V signals: never 24 V on them; twisted, away from 230 V and motor cables. GX16 (round connector): slide the shell and heat-shrink on before soldering. NPN NO sensors: brown +24 V, blue 0 V.								
90	☐	**210**	E-stop 4	A11 X7	1 yellow	**dk blue**	**2.00**	fer / fer	E-stop signal to the controller; door loop ⚠ **§ 7.1 (dead tests): X7 (E-stop input) opens with the E-stop pushed**
91	☐	**224**	Green lamp -	A11 Y6	1 yellow	**dk blue**	**2.00**	fer / fer	Work; door loop
92	☐	**226**	Red lamp -	A11 Y5	1 yellow	**dk blue**	**2.05**	fer / fer	Alarm; door loop
93	☐	**301**	A11 X_5V	DMA860H-X PUL+	1 yellow tw	**dk blue**	**0.85**	fer / 2×+304	
94	☐	**302**	A11 X_PUL	DMA860H-X PUL-	1 yellow tw	**dk blue**	**0.85**	fer / fer	
95	☐	**303**	A11 X_DIR	DMA860H-X DIR-	1 yellow tw	**dk blue**	**0.95**	fer / fer	
96	☐	**304**	DMA860H-X PUL+	DMA860H-X DIR+	1 yellow	**dk blue**	**0.20**	2×+301 / fer	Jumper
97	☐	**305**	A11 Y_5V	DMA860H-Y PUL+	1 yellow tw	**dk blue**	**1.05**	fer / 2×+308	
98	☐	**306**	A11 Y_PUL	DMA860H-Y PUL-	1 yellow tw	**dk blue**	**1.05**	fer / fer	
99	☐	**307**	A11 Y_DIR	DMA860H-Y DIR-	1 yellow tw	**dk blue**	**1.05**	fer / fer	
100	☐	**308**	DMA860H-Y PUL+	DMA860H-Y DIR+	1 yellow	**dk blue**	**0.20**	2×+305 / fer	Jumper
101	☐	**309**	A11 Z_5V	DMA860H-Z PUL+	1 yellow tw	**dk blue**	**1.15**	fer / 2×+312	
102	☐	**310**	A11 Z_PUL	DMA860H-Z PUL-	1 yellow tw	**dk blue**	**1.25**	fer / fer	
103	☐	**311**	A11 Z_DIR	DMA860H-Z DIR-	1 yellow tw	**dk blue**	**1.25**	fer / fer	
104	☐	**312**	DMA860H-Z PUL+	DMA860H-Z DIR+	1 yellow	**dk blue**	**0.20**	2×+309 / fer	Jumper
105	☐	**315**	DMA860H-X ALM+	A11 X5	1 yellow	**dk blue**	**0.75**	2×+317 / fer	Drive alarm (wire-OR)
106	☐	**317**	DMA860H-Y ALM+	DMA860H-X ALM+	1 yellow	**dk blue**	**0.55**	2×+319 / 2×+315	NEW: X5 had 3 conductors
107	☐	**319**	DMA860H-Z ALM+	DMA860H-Y ALM+	1 yellow	**dk blue**	**0.55**	fer / 2×+317	
108	☐	**325**	A11 Y1	VFD X1	1 yellow	**dk blue**	**0.70**	fer / fer	Spindle run (forward)
109	☐	**327**	A11 Y2	VFD X2	1 yellow	**dk blue**	**0.70**	fer / fer	Multi-speed bit 1
110	☐	**328**	A11 Y3	VFD X3	1 yellow	**dk blue**	**0.75**	fer / fer	Multi-speed bit 2
111	☐	**329**	A11 Y4	VFD X4	1 yellow	**dk blue**	**0.75**	fer / fer	Multi-speed bit 3
112	☐	**330**	X-LIM 1	A11 X1	1 yellow	**dk blue**	**0.70**	solder / fer	X home (X–)
113	☐	**331**	Y-LIM 1	A11 X2	1 yellow	**dk blue**	**0.75**	solder / fer	Y home (Y–)
114	☐	**332**	Z-LIM 2	A11 X3	1 yellow	**dk blue**	**0.80**	solder / fer	Z home (Z+)
115	☐	**334**	X-LIM 2	Y-LIM 2	1 yellow	**dk blue**	**0.40**	solder / solder	Y+ limit (loop)
116	☐	**335**	Y-LIM 2	Z-LIM 1	1 yellow	**dk blue**	**0.40**	solder / solder	Z– limit (loop)
117	☐	**336**	X-LIM 2	A11 X6	1 yellow	**dk blue**	**0.75**	solder / fer	Hard limit (wire-OR)
118	☐	**345**	Tool setter 1	A11 X4	1 yellow	**dk blue**	**0.85**	solder / fer	Tool-setter touch
119	☐	**347**	Tool setter 3	E-stop 3	1 yellow	**dk blue**	**2.35**	solder / fer	Tool-setter over-travel (NC) in series with the E-stop NC2 → A11 X7 (door loop); door loop ⚠ **§ 7.1 (dead tests): X7 (E-stop input) opens with over-travel pushed**
120	**Multi-core cables – 4, cut once per cable to the longest core**								
121	⚠ **CHECK FIRST:** Motor cables: Ω-check the coil pairs at the motor before soldering (A+ / A– one coil, B+ / B– the other); never plug or unplug a motor with its drive powered; LiYCY braid to the plate clamp at the drive end. W1 (spindle cable): 360° shield at the HDC-HA (spindle connector) hood and the VFD (spindle drive) clamp; U / V / W / PE only.								
122	☐		**Motor cable X: LiYCY 4 × 1.0 shielded**				**1.15**	DMA860H-X A+ → XM-X 1 · DMA860H-X A- → XM-X 2 · DMA860H-X B+ → XM-X 3 · DMA860H-X B- → XM-X 4	
123	☐		**Motor cable Y: LiYCY 4 × 1.0 shielded**				**1.00**	DMA860H-Y A+ → XM-Y 1 · DMA860H-Y A- → XM-Y 2 · DMA860H-Y B+ → XM-Y 3 · DMA860H-Y B- → XM-Y 4	
124	☐		**Motor cable Z: LiYCY 4 × 1.0 shielded**				**0.95**	DMA860H-Z A+ → XM-Z 1 · DMA860H-Z A- → XM-Z 2 · DMA860H-Z B+ → XM-Z 3 · DMA860H-Z B- → XM-Z 4	
125	☐		**Spindle cable W1: 4G1.5 shielded**				**1.05**	123: VFD U → HDC-HA 1 · 124: VFD V → HDC-HA 2 · 125: VFD W → HDC-HA 3 · 126: VFD PE → HDC-HA PE · 127: HDC-HA SH → VFD PEb	
126	**Before power**								
127	⚠ **Before first power** (§ 7.1, QF0 (main breaker) OFF, Ω meter): Z1 (EMI filter) LOUT ↔ NOUT open or high · 230 V (Z1 LOUT) ↔ 0 V, +24 V and PE: open · 148: 0 V ↔ PE beeps · G1 (48 V PSU) + / – and G2 (24 V PSU) + / –: not 0 Ω · A11 (controller) X / Y / Z 5V ↔ G2 +V: open · A11 PUL ↔ G1 +V: open · ON / OFF / E-stop chain and K1 (latch relay) test lever as § 7.1 (dead tests).								
128	⚠ **Power-up in order** (§ 7.2): QF1–QF3 OFF, 131 / 166 / 168 off → QF0 (main breaker) ON, hands off the plate → F2 (signal fuse) out 23–25 V DC, then connect the A11 (controller) → ON holds K1 (latch relay), E-stop drops it → QF0 OFF, fit 131, ON → XL (switched L busbar) ↔ XN (N busbar) ≈ 230 V AC → QF2 (48 V PSU breaker) ON (drives off): G1 (48 V PSU) 46–50 V → fit 150–155, jog each axis slowly → VFD (spindle drive) COM ↔ G2 (24 V PSU) –V < 1 V → spindle.								
129	**Open:** 2.5 mm² G/Y short by 3.95 m – buy 5 m (or 1.5 mm² G/Y for PE 145, 146, 148, 149) · braking resistor leads 128 / 129 are 1 mm² PVC (owner's choice) · not listed: 12 conductors outside the box (motor and sensor cables to the machine) · source A11_Wire_Lengths_rev07.csv (player -a11lengths), regenerate with make_wire_list.py								